



Maxim Bejenari

Passport: ... | **Date of birth:** 18/05/2005 | **Place of birth:** Balti, Moldova |

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<https://github.com/maxonbejenari> | **Address:** Bacau, Romania (Home)

ABOUT ME

I am a motivated and curious person, always eager to learn and adapt to new challenges. I value collaboration, creativity, and continuous improvement.

EDUCATION AND TRAINING

01/09/2021 – 31/05/2024 Balti, Moldova

HIGH SCHOOL DIPLOMA THEORETICAL LYCEUM "LUCIAN BLAGA"

Field of study Mathematics, Physics, Biology, Chemistry

01/10/2024 – CURRENT Bacau, Romania

BACHELOR "Vasile Alecsandri" University of Bacau, Faculty of Engineering

Website <https://www.ub.ro/> | **Field of study** Mechatronics Engineering

01/03/2026 – 01/07/2026 Istria, Croatia

ERASMUS+ ISTRIAN UNIVERSITY OF APPLIED SCIENCES

Website <https://www.iv.hr/en/istrian-university-of-applied-sciences/> | **Field of study** Mechatronics Engineering

LANGUAGE SKILLS

Mother tongue(s): **ROMANIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
RUSSIAN	C1	C1	C1	C1	B2
ENGLISH	B2	B2	B2	B2	B2
GERMAN	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

software engineering | robotics engineering | C++ | Python (computer programming) | Git | Linux | SQL | ROS2 | Electrical Engineering | Mechanical Engineering | CAD | PLC | Arduino | Raspberry Pi | MATLAB | computer vision | OpenCV

PROJECTS

Mobile App Development Project – Tekwill National Contest, Moldova

Contributed to designing and developing a Java-based mobile application that provides teachers with a digital timetable management tool.

Link <https://github.com/maxonbejenari/E-Orar>

Tekwill National Contest – Algorithm Design (C++)

Participated in a national competition focused on designing and solving algorithms using C++.

01/02/2026 – CURRENT

The Robot Vacuum Cleaner Simulation Prototype

This project is a prototype autonomous vacuum cleaner robot developed entirely in simulation. The robot was designed in Onshape, converted to URDF using onshape-to-robot, simulated in Gazebo, and equipped with:

- SLAM Toolbox for real-time mapping and localization
- NAV2 for autonomous point-to-point navigation
- YOLOv8 for real-time object detection, trained on custom classes (e.g. cube detection)

The goal is to simulate the full software stack of a commercial robot vacuum cleaner — from sensor fusion and mapping to coverage path planning and autonomous navigation.

Link https://github.com/maxonbejenari/Robot_Vacuum_Cleaner_ROS2

Strength Analysis of Pressure Vessels (Erasmus+ Project) | Istrian University of Applied Sciences

- Conducted a detailed strength analysis of spherical pressure vessels under various operating temperatures (up to 200°C), in accordance with the AD 2000-Merkblatt safety standards.
- Calculated the minimum wall thickness and structural safety factor for various industrial steel grades (EN 10025 S355J2, NIOVAL 47).
- Validated the analytical calculations through Finite Element Analysis (FEA) on a 3D model, achieving a precise correlation between the numerical simulations and theoretical results.

Link <https://drive.google.com/file/d/1-iSkjDhbVIWjR8x-nVtzHDBpjaPq-p/view?usp=sharing>